

UNIT-3 | Lecture 1

Relational Algebra, Tuple & Domain Relational Calculus

Coverage: Hours 1–2

Learning Objectives

Understand relational algebra operators and the difference between procedural and non-procedural query languages.

Understand TRC and DRC notation.

Relational Algebra

Relational algebra is a procedural query language that applies operations to relations to produce new relations.

Selection (σ): selects rows satisfying a condition. Example: σ Branch='CSE'(Student).

Projection (π): selects required columns. Example: π Name,Branch(Student).

Union ($\cup$): combines tuples from union-compatible relations.

Difference ($-$): returns tuples in the first relation but not the second.

Cartesian Product ($\times$): combines every tuple of two relations.

Rename (ρ): renames a relation/attributes. **Join** combines related tuples based on a condition.

Tuple Relational Calculus (TRC)

TRC is **non-procedural**: it specifies *what* tuples are required, not the sequence of operations.

General form: $\{ t \mid P(t) \}$, where t is a tuple variable and $P(t)$ is a predicate.

Domain Relational Calculus (DRC)

DRC is also **non-procedural**, but variables represent individual attribute/domain values rather than complete tuples.

General form: $\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$.

Comparison

Relational Algebra	TRC / DRC
Procedural — how to obtain result	Non-procedural — what result is required
Uses relational operators	Uses logical predicates

Key Takeaway

Relational algebra is procedural, while TRC and DRC express queries declaratively.